THOMAS TO

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PROFESSIONAL SUMMARY

Full Stack AI Engineer with 7+ years of professional experience building end-to-end systems across software development and machine learning lifecycle. Contributes actively to open-source projects including OpenXLA, JAX, and Kubernetes, supporting TPU and GPU-accelerated computation across distributed training and inference workloads. Collaborates with technical and non-technical stakeholders to evaluate engineering trade-offs and deliver measurable business outcomes. Featured by International Business Times as a thought leader on AI-enabled drug discovery, demystifying AI/ML for the general public leveraging my credibility.

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, JavaScript, TypeScript, Google Apps Script, R, HTML, CSS, MATLAB
- **Web and APIs:** React, Next.js, Vue.js, Node.js, RESTful APIs, FastAPI, Tailwind CSS, Streamlit, Mermaid.js
- **AI and ML:** Generative AI, Large Language Models (OpenAI GPT, Anthropic Claude, Google Gemini, Snowflake Cortex), Retrieval-Augmented Generation, Prompt Engineering, Fine-Tuning, LangChain, LangGraph, LangSmith, MCP, n8n, Vector Databases, Embedding Models, Hugging Face, Model Evaluation, Guardrails, PyTorch, NumPy, pandas, OpenCV, TensorBoard, RDKit, Molecular Dynamics & Docking, Chemistry & Cheminformatics, pyRosetta, PyMOL, FoldIT, Rosetta, OpenFold3, Rosetta Foundry
- **Cloud and DevOps:** GCP, AWS, Docker, CI/CD, MLOps, Containerization, Kubernetes, Vercel, GitHub Actions, Model Monitoring, Infrastructure as Code, Observability, OpenXLA, JAX, CuPy
- **Data Management:** Snowflake, MongoDB, PostgreSQL, dbt, Fivetran, SAP S/4HANA, ETL/ELT Pipelines, Data Warehousing, Data Modeling, Tableau, Sigma

PROFESSIONAL EXPERIENCE

Founding AI Engineer | Open Source | Oakland, CA | *Dec 2017 – Present*

- Contributes to OpenXLA, JAX, and CuPy open-source projects supporting TPU and GPU-accelerated computation, and Kubernetes for large-scale server orchestration applied to ml-drug-discovery, OpenFold3, and Rosetta Foundry for computational protein modeling.
- Architected a production Retrieval-Augmented Generation (RAG) AI agent with Large Language Model (LLM) output validation and guardrails on Next.js and React using Typescript. Shipped 0-to-1 from system design through open-source distribution, adopted by 10+ engineers for deployment.
- Launched an agentic AI pipeline integrating LLM orchestration with prompt engineering, output guardrails, and exploratory data analysis, generating 7 automated posts per week and saving 5+ hours per week through CI/CD automation via GitHub Actions.
- Slides-as-Code reducing presentation creation time by 75%.
- Built an AI/ML arXiv research agent and newsletter serving engineers and academics, achieving a 100% free-forever production service by deploying a 4-bit quantized 7B model on Oracle Cloud Infrastructure to eliminate token costs, and engineering an automated fallback to Google AI Studio to handle volume spikes and multimodal inputs.

Founding Fullstack Engineer | Canventa Life Sciences | Emeryville, CA | *Jan 2023 – Present*

- Architected a revenue optimization system integrating a predictive machine learning model with a RAG AI agent on Snowflake, enabling non-technical stakeholders to query revenue data via natural language. Reduced decision cycles from 3+ hours to under 10 minutes, saving 500+ hours annually.
- Achieved 95%+ accuracy digitizing 5,000+ handwritten laboratory documents by fine-tuning a document-understanding model (Snowflake Arctic-TILT) with custom embeddings and annotated training data. Saved 1,000+ hours of manual transcription through systematic model evaluation.
- Engineered end-to-end ETL pipelines with Python, SQL, and Google Apps Script ingesting from 3+ enterprise systems (SAP, CRIO, FreezerPro, Google Workspaces) into Snowflake fact and dimension tables. Reduced data processing time from days to minutes, saving 20+ hours per week.
- Reduced ad-hoc data requests by 70% and recovered 30+ production hours per week by building data validation pipelines with Python and dbt on Snowflake, with Streamlit for self-serve querying and Tableau for enterprise dashboards.
- Reduced $200K in material waste and prevented a $2M inventory stockout by engineering data-driven demand forecasting and inventory optimization processes.

Software Engineer | Genentech | South San Francisco, CA | *Jun 2022 – Jan 2023*

- Engineered full-stack applications with Python REST API, React, Vue.js, and Node.js, streamlining workflows for 5+ scientific teams and saving 15+ hours per week.
- Accelerated feature delivery from 4+ weeks to under 1 week for internal tooling by incorporating stakeholder feedback into iterative development cycles across cross-functional teams.

Process Engineer | Genentech | Vacaville, CA | *Jun 2021 – Jun 2022*

- Reduced data processing time by 99% (weeks to minutes) by developing automated data pipelines with Python, R, and SQL for parallel execution and batch processing optimization.
- Digitized manual workflows into structured databases and reporting dashboards serving 5+ cross-functional teams, saving 10+ hours per week and enabling real-time visibility into production metrics.

Research Engineer | UC Davis (Nandi, McDonald, Wan, Siegel Labs) | Davis, CA | *Sep 2019 – Jun 2021*

- Reduced manufacturing costs by $63.2M in modeled scenarios by designing computational optimization models with Python (NumPy, pandas), numerical algorithms, and parallel execution for process engineering.
- Built Python computer vision pipelines with OpenCV for automated microscopy image analysis, processing 10,000+ images in hours versus weeks of manual counting for quantitative measurement.
- Published 3 novel protein variants for biomanufacturing by designing computational protein models with molecular simulation frameworks (pyRosetta, PyMOL), validating through wet-lab testing and iterative optimization.

Teaching Assistant | University of California, Davis | Davis, CA | *Sep 2019 – Jun 2022*

- Migrated legacy MATLAB codebase to Python3 while developing supplemental coding exercises and debugging workshops covering data preprocessing, feature engineering, and iterative model development.

Laboratory Coordinator | Laney College | Oakland, CA | *Aug 2017 – Jun 2019*

- Designed Python programming and data analysis experimentation into STEM coursework reaching 50+ students per semester through hands-on exercises in visualization and computational modeling.

PROJECTS

Real-Time WebSocket Puzzle Agent | *Jun 2026*

- Engineered an autonomous Node.js WebSocket agent with regex-first dispatch resolving 6 checkpoint types in sub-millisecond time against a 4-second server deadline, integrating a custom recursive-descent math parser, Wikipedia REST API, and Google Gemini 2.5 Flash function-calling fallback, validated by 107 test cases.

Embedded Edge AI on Raspberry Pi

- Deployed NVIDIA Alphamayo VLM+A models on Raspberry Pi for real-time edge inference at sub-200ms latency with zero cloud dependency, applying model quantization and Docker containerization.

Agentic Video Editing Pipeline

- Shipped a 0-to-1 open-source screen recording and editing platform reducing video production time by 80%, architecting an agentic pipeline with FFmpeg for automated video processing, Google Chrip 3 for speech-to-text transcription, and YouTube OAuth API for SEO-optimized publishing.

Document (ETL) Form Chrome Extension | *Jan 2026*

- Built a Chrome extension as a service paired with a Vercel-deployed React/Next.js frontend, implementing hashing, encryption, and Chrome security permissions for targeted HTML parsing across sheet URLs, in TypeScript with Google ADK and Anthropic Claude Code. Eliminating manual form filling for law firm documents and passport records by shipping an agentic ETL pipeline processing uploads through in-memory storage and auto-populating target ingestion forms, reducing a multi-hour manual process to seconds.

MEDIA & PUBLICATIONS

"Thomas To on How AI-Enabled Biology is Advancing Drug Discovery and Scientific Innovation" | International Business Times | *Jun 2026*

- Featured as a thought leader on AI in scientific research, discussing AI as a support tool for scientific decision-making rather than a replacement for empirical biology and scientific expertise. Made the case for specialized systems trained on scientific data over general conversational AI.

"Design to Data for mutants of β-glucosidase B from Paenibacillus polymyxa: L171G, L171V and L171W" | McNair Scholars Program, UC Davis Genome Center | Mentored by Dr. Justin Siegel and Ashley Vater

- Designed and characterized three novel single-point enzyme mutants (L171G, L171V, L171W) using Rosetta and FoldIT computational modeling paired with Kunkel mutagenesis. Contributed kinetic and thermal stability data to the Design2Data (D2D) protein engineering database.

EDUCATION

Bachelor of Science, Biochemical Engineering | University of California, Davis | Davis, CA | *Sep 2019 – Jun 2022*

CERTIFICATIONS, AWARDS, & LEADERSHIP

Honors: AvenueE Engineering Leadership Program, McNair Scholars TRIO Program Fellow, Genentech Leadership Exchange

Relevant Coursework: Advanced Machine Learning with Agentic AI, AWS Cloud Technical Essentials, Data Engineering Foundations

Leadership:

- Produce and publish MLOps-focused educational content on YouTube and LinkedIn through an automated CI/CD publishing pipeline, driving developer community engagement and technical thought leadership.
- Serve as Student Outreach Ambassador supporting 100,000+ community college transfer students by organizing peer-to-peer mentorship programs, transfer outreach initiatives, and cross-institutional community building.
- Engage across engineering, pharmaceutical, and AI communities through active membership in AIChE, ISPE, and the Rosetta protein engineering community, and as a panelist in the Ipsos AI Insights Community and inaugural Unintentional Consequences of Technology (UCOT) conference.
- Organize and contribute to collaborative workshops and learning initiatives including Interview Kickstart bootcamp peer sessions, Databricks weekly seminars, and enterprise analytics community forums to support data engineering and analytics practitioners.
- Coach students ages 3 to adult in Brazilian jiu-jitsu, delivering structured athletic training to build discipline, resilience, and community belonging in Oakland.